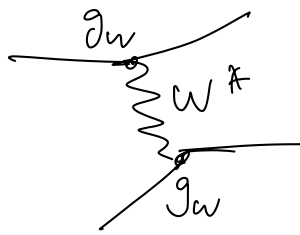


Weak Interaction

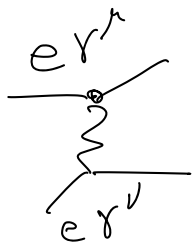
~~Goldberger experiment~~

~~WW experiment~~

$\Rightarrow$ weak interactions include chirality of particles



$\frac{g_W^2}{q^2 - m_W^2} (g^{\mu\nu} - \frac{q^\mu q^\nu}{m_W^2})$ massive vector boson propagator.



QED

$$\mathcal{M}_{QED} \sim \frac{e^2}{4q} (\bar{\psi} \gamma^\mu \psi) \frac{g_{\mu\nu}}{q^2} (\bar{\psi} \gamma^\nu \psi)$$

$$\sim \frac{\alpha}{q^2} \underline{J}_\mu \cdot \underline{J}^\mu = \frac{\alpha}{q^2} \underline{J} \cdot \underline{J}$$

$$\underline{J}^\mu = \underline{J} = (\bar{\psi} \gamma^0 \psi, \bar{\psi} \gamma^i \psi)$$

4-current of QED.

$$\mathcal{M}_{QED} \sim \frac{\alpha}{q^2} \underline{J} \cdot \underline{J} \quad \text{current-current interac.}$$

$\underline{J}$ is a four vector $\Rightarrow$ under Lorentz transf.

$$\underline{J} = (J^0, \vec{J})$$

$\underline{J}$ follows the rules of $\underline{x}, \underline{p}$

~~$\mathcal{M}_{QED} = ?$~~

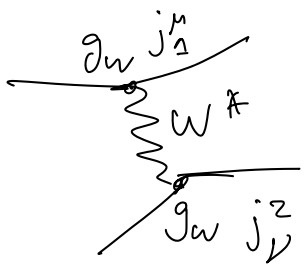
$$\begin{aligned} \underline{P} \underline{J} &= \underline{P} (\bar{\psi} \gamma^0 \psi, \bar{\psi} \gamma^i \psi) = (\bar{\psi} \gamma^0 \psi, -\bar{\psi} \gamma^i \psi) \\ &= (j^0, -\vec{J}) \end{aligned}$$

$\bar{\psi} \gamma^\mu \psi$ is a vector under Lorentz transf

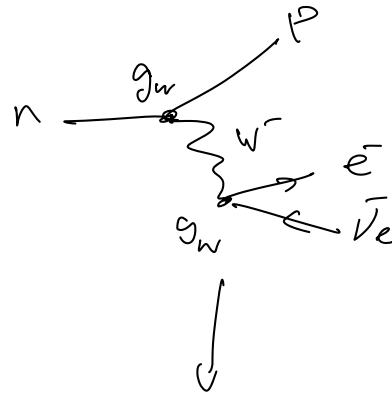
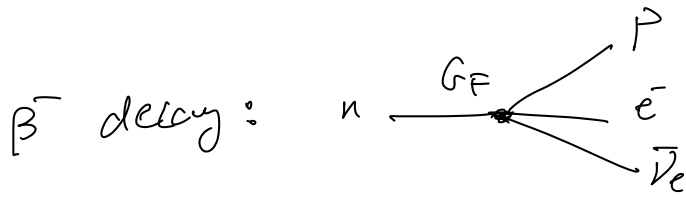
$$\mathcal{M}_{QED} = \frac{\alpha}{q^2} \underline{J} \cdot \underline{J} = \frac{\alpha}{q^2} (j_1^0 j_2^0 - \vec{J}_1 \cdot \vec{J}_2)$$

$$\underline{P} \mathcal{M}_{QED} = \frac{\alpha}{q^2} (j_1^0 j_2^0 - (-\vec{J}_1) \cdot (-\vec{J}_2)) = \mathcal{M}_{QED}$$

$\Rightarrow$ QED is invariant under $\underline{P}$



$$\frac{g_w^2}{q^2 - m_w^2} (g^{\mu\nu} - \gamma^\mu \gamma^\nu / m_w^2) j_1^\mu j_2^\nu$$



In similarity w/ QED:

$$M_{\text{weak}} = G_F \underline{J}_{\text{up}} \cdot \underline{J}_{\text{ev}}$$

$$M_{\text{weak}} = \frac{\alpha}{q^2 - m_w^2} \underline{J}_{\text{up}} \cdot \underline{J}_{\text{ev}}$$

$$\underline{J}_{\text{up}} = \bar{\psi}_n \gamma^\mu \psi_p \quad \underline{J}_{\text{ev}} = \bar{\psi}_{\nu_e} \gamma^\mu \psi_e \quad ?$$

same Lorentz structure of QED for weak interactions.

If $M_{\text{weak}} = M_{\text{weak}}$. $\Rightarrow$ no space for ~~$\not{A}$~~
 Contradicted by Wu experiment.
 also not working for ~~$\not{A}$~~ seen by Goldhaber.

we need to incorporate ~~$\not{A}$~~ and ~~$\not{S}$~~

$\mathbb{P} \vec{V} \rightarrow -\vec{V}$		$\bar{\psi} \gamma^\mu \psi$
$\mathbb{P} S \rightarrow S$		$\bar{\psi} \psi$
$\mathbb{P} \vec{A} \rightarrow \vec{A}$	pseudo-vector axial vector	$\bar{\psi} \gamma^\mu \gamma^5 \psi$
$\mathbb{P} PS \rightarrow -PS$	pseudo-scalar	$\bar{\psi} \gamma^5 \psi$

$$\underline{J}_{\text{weak}} = ? \quad \Rightarrow \quad M_{\text{weak}} \text{ does not conserve } \mathbb{P}$$

$$\underline{J}_{\text{weak}} = \bar{\psi} (C_V \gamma^\mu + C_A \gamma^5 \gamma^\mu + C_T \sigma^{\mu\nu}) \psi$$

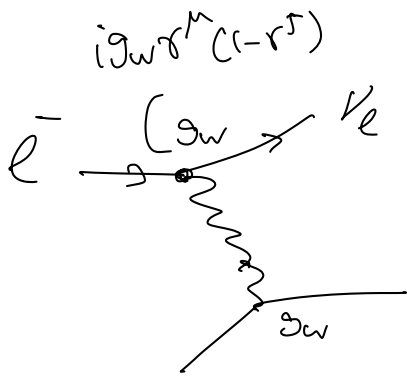
$\sigma^{\mu\nu} = \frac{i}{2} [\gamma^\mu, \gamma^\nu]$ tensor bilinear

most general form for $\underline{j}_{weak}$ allows $\cancel{A}$ and $\cancel{S}$

$$\mathcal{M}_{weak} \propto (1) Q^2 + (1) C_A^2 + (-1) G^2 + (-1) C_V C_A + (-1) C_V C_T + \dots$$

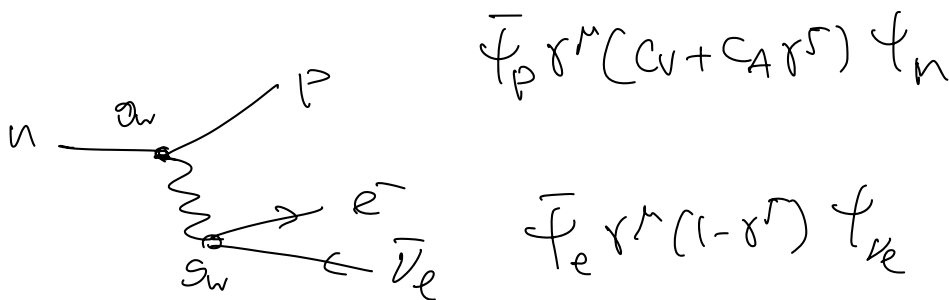
From experimental results $\Rightarrow j_{weak} = \bar{\psi} \gamma^\mu (1 - \gamma^5) \psi$
 for leptons
 $= \bar{\psi} \gamma^\mu \psi - \bar{\psi} \gamma^\mu \gamma^5 \psi$
 $= V - A$

$\Rightarrow$ V-A theory for weak interactions



$$\mathcal{M}_{weak} = \frac{g_w^2}{q^2 - m_W^2} (\bar{\psi} \gamma^\mu (1 - \gamma^5) \psi) \times (\bar{\psi} \gamma_\mu (1 - \gamma^5) \psi)$$

V-A valid for leptons and quarks. (fundamental fermions).



$$\bar{\psi}_p \gamma^\mu (C_V + C_A \gamma^5) \psi_n$$

$$\bar{\psi}_e \gamma^\mu (1 - \gamma^5) \psi_{\bar{\nu}_e}$$

C_V, C_A measured experimentally.

Experimental verification of V-A theory?

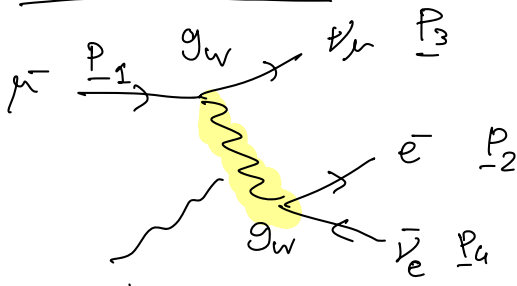
- $\mu^- \rightarrow \nu_\mu e^- \bar{\nu}_e$ β decay
- $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$ pure leptonic decay
- $\pi^- \rightarrow e^- \bar{\nu}_e / \mu^- \bar{\nu}_\mu$

1) verify that V-A is correct.

2) is g_w the same for $e^-, \mu^-, \pi^-, p, n, \dots$

Muon decay

Griffiths 9.2 (Goldberger ex. 6.7)



$$\underline{q} = \underline{p}_1 - \underline{p}_3$$

$$\frac{g_{\mu\nu}}{q^2 - m_W^2}$$

$$q^2 \sim m_\mu^2 \ll m_W^2 = \frac{g_{\mu\nu}}{q^2 - m_W^2} \approx \frac{g_{\mu\nu}}{m_W^2}$$

$$\mathcal{M} = \left(\frac{g_W^2}{q^2 - m_W^2} \right) \times [\bar{u}(p_3) \gamma^\mu (1 - \gamma^5) u(p_1)] [\bar{u}(p_4) \gamma^\mu (1 - \gamma^5) u(p_2)]$$

$\rightarrow G_F^2$

$$\Gamma(\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu) \propto \langle |\mathcal{M}|^2 \rangle \times (\text{phase space})$$

$$\langle |\mathcal{M}|^2 \rangle = \sum_{\text{spins final state}} \sum_{\text{average spin init. state}} |\mathcal{M}|^2$$

Init. State: $\mu^- \xrightarrow{\Leftarrow} \xrightarrow{\Rightarrow}$ 2 init. contr.

Final State:

- $\nu_\mu \xrightarrow{\Leftarrow} 1 \text{ state}$
- $\bar{\nu}_e \xrightarrow{\Rightarrow} 1 \text{ state}$
- $e^- \xrightarrow{\Leftarrow} \xrightarrow{\Rightarrow} 2 \text{ states.}$

phase space: $\rho = \frac{d^3 p_{\nu_e}}{(2\pi)^3} \frac{d^3 p_e}{(2\pi)^3} \frac{d^3 p_{\nu_\mu}}{(2\pi)^3} \delta(\underline{p}_\mu - \underline{p}_e - \underline{p}_{\bar{\nu}_e} - \underline{p}_{\nu_\mu})$

$$\Gamma \propto |\mathcal{M}|^2 \times \rho(\text{phase space})$$

$$d^3 p_e = 2\pi d\cos\theta_e d\phi_e$$

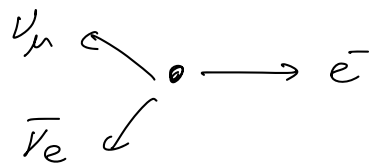
$$\frac{d^3 p_e}{d^3 p_e} = 1$$

$$|\mathcal{M}| \propto G_F = \frac{g_W^2}{m_W^2}$$

$$[\mathcal{M}] = [E]^2$$

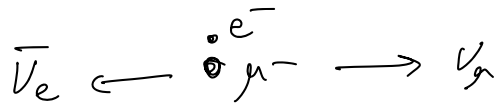
$$[\rho] = E^3 E^{-4} = E^{-1}$$

In rest. solidal with μ^- .



Generic Config.

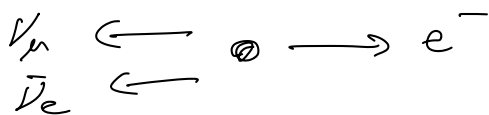
Min energy for e^-



$$E_{\nu_e} = E_{\nu_\mu} = \frac{1}{2} m_\mu = 53 \text{ MeV}$$

$$E_e = m_e \quad T_e \approx 0$$

Max energy for e^-



$$E_e = \frac{1}{2} m_\mu$$

$$E_e \in [m_e, m_\mu/2]$$

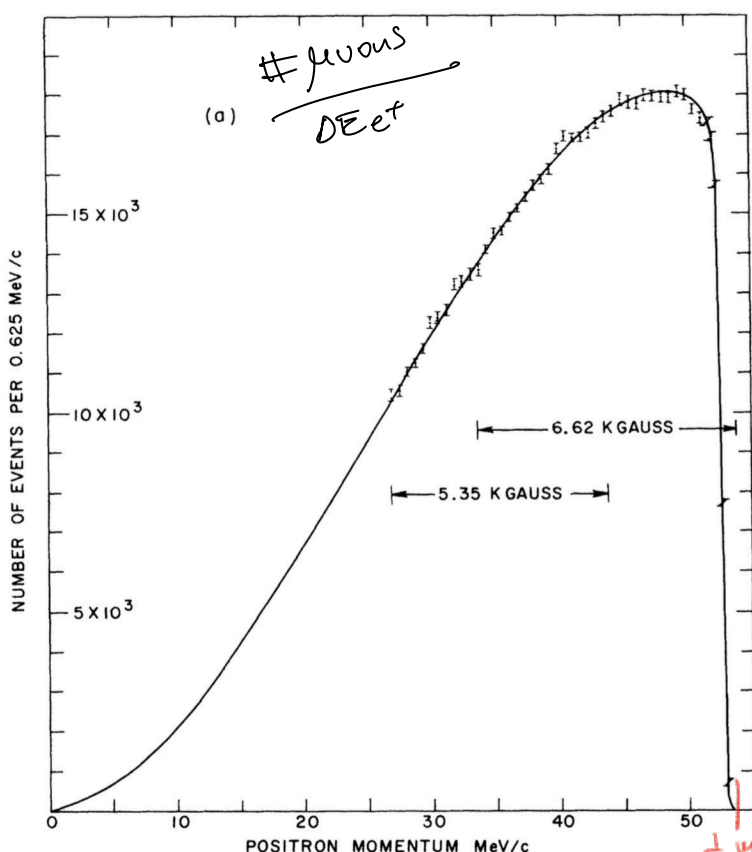
From dimensional analysis

$$\Gamma_\mu \propto G_F^2 m_\mu^5$$

from phase space

$$\Gamma_\mu = \frac{1}{192 \pi^3} G_F^2 m_\mu^5$$

(from calculation w/ spinors)



$$\text{bin width} = 0.625 \text{ MeV}$$

produce many μ^+ .

$\rightarrow$ more than decay.

$\rightarrow$ measure E_{e^+}

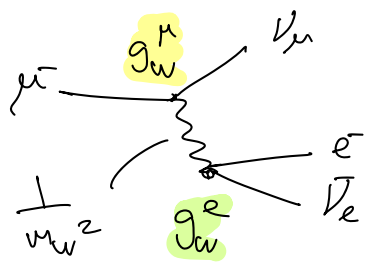
$$\rightarrow \frac{d\Gamma}{dE_{e^+}} \equiv \frac{\# \mu \text{ decay}}{\Delta E_{e^+}}$$

$$\mu^+ \rightarrow e^+ \nu_e \bar{\nu}_\mu$$

successful comparison of prediction with data

$$g_w^\mu \equiv g_w^e$$

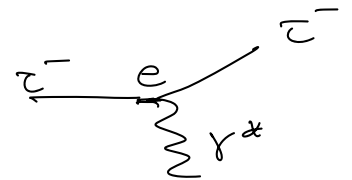
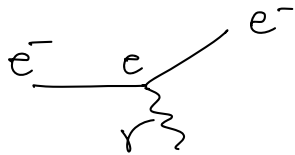
Lepton Flavor Universality of weak interactions



$$\Gamma \sim \underbrace{\left(\frac{g_w^\mu}{m_W}\right)^2}_{G_F} \underbrace{\left(\frac{g_w^e}{m_W}\right)^2}_{G_F} m_\mu^5$$

Experimental evidence for $g_w^e = g_w^\mu = ?$

Go back to QED



$$P = q B R$$

measure $P \Rightarrow$ measure q

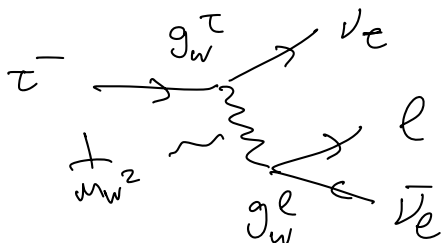
if $g_w^\mu \neq g_w^e \Rightarrow \Gamma \propto \left(\frac{g_w^e}{m_W}\right)^2 \left(\frac{g_w^\mu}{m_W}\right)^2 m_\mu^5$

if $g_w^\mu = g_w^e \Rightarrow$ weak int. has lepton flavor universality

what about τ^- ?

$$g_w^e \equiv g_w^\mu \equiv g_w^\tau ?$$

τ^- decays (weak)



$$m_\tau = 1.78 \text{ GeV} \ll m_W$$

$$m_\mu = 0.106 \text{ GeV}$$

$$m_e = 0.511 \text{ MeV}$$

$$\tau^- \rightarrow \mu^- \bar{\nu}_\mu \nu_\tau \quad \Gamma_\mu \sim \left(\frac{g_w^\tau}{m_W}\right)^2 \left(\frac{g_w^\mu}{m_W}\right)^2 \times \text{phase space } (\tau \rightarrow \mu)$$

$$\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau \quad \Gamma_e \sim \left(\frac{g_w^\tau}{m_W}\right)^2 \left(\frac{g_w^e}{m_W}\right)^2 \times \text{phase space } (\tau \rightarrow e)$$

$$Q(\tau \rightarrow \mu) = m_\tau - m_\mu - \cancel{m_\mu} - \cancel{m_\mu} = m_\tau - m_\mu = 1.78 - 0.11 = 1.67 \text{ GeV}$$

$$Q(\tau \rightarrow e) = m_\tau - m_e \approx m_\tau = 1.78 \text{ GeV} > Q_\mu.$$

$$\Gamma(\tau \rightarrow \mu \bar{\nu}) \propto N(\tau \rightarrow \mu)$$

$$\Gamma(\tau \rightarrow e \bar{\nu}) \propto N(\tau \rightarrow e)$$

$$\frac{\#(\tau \rightarrow \mu)}{\#(\tau \rightarrow e)} = \frac{\Gamma(\tau \rightarrow \mu)}{\Gamma(\tau \rightarrow e)} = \frac{(g_w^\mu)^2}{(g_w^e)^2} \frac{\rho(\tau \rightarrow \mu)}{\rho(\tau \rightarrow e)}$$

↕
events. for e/μ.
measure.

Computed analytically

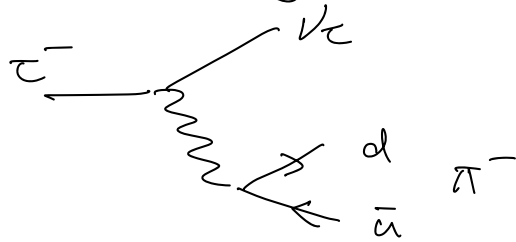
$$\Rightarrow \frac{g_w^\mu}{g_w^e} \approx 1.001 \pm 0.002 \Rightarrow g_w^e \equiv g_w^\mu$$

same coupling constant for e, μ

what about g_w^τ ?

$\Gamma_{\text{tot}}(\tau)$

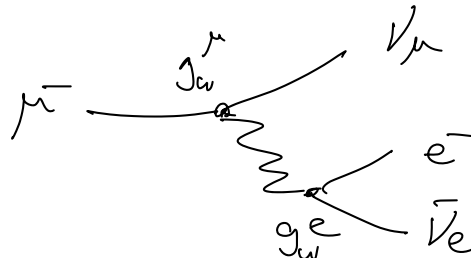
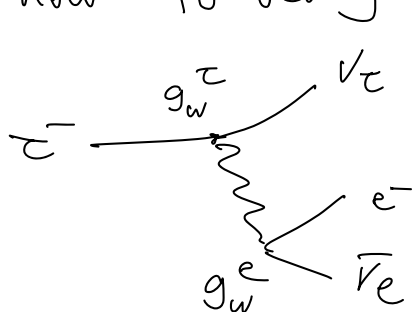
τ can decay to hadrons.



$$\tau^- \rightarrow \pi^- \nu_\tau$$

$$Q = m_\tau - m_\pi \approx Q_\mu$$

how to verify $g_w^\tau = g_w^\mu = g_w^e$?



$$\frac{\Gamma(\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau)}{\Gamma(\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu)} = \left(\frac{g_w^\tau}{g_w^\mu} \right)^2 \times \frac{\rho(\tau \rightarrow e)}{\rho(\mu \rightarrow e)}$$

$$\frac{\#(\tau \rightarrow e)}{\#(\mu \rightarrow e)} \Rightarrow \frac{g_w^\mu}{g_w^\tau} = 1.001 \pm 0.003$$

Computed analytically.

τ^- DECAY MODES	Fraction (Γ_i/Γ)	Scale factor/ Confidence level	p (MeV/c)
Modes with one charged particle			
particle ⁻ ≥ 0 neutrals ≥ 0 $K^0 \nu_\tau$ ("1-prong")	(85.24 ± 0.06) %		-
particle ⁻ ≥ 0 neutrals ≥ 0 $K_L^0 \nu_\tau$	(84.58 ± 0.06) %		-
$\mu^- \bar{\nu}_\mu \nu_\tau$	[g] (17.39 ± 0.04) %		885
$\mu^- \bar{\nu}_\mu \nu_\tau \gamma$	[e] (3.67 ± 0.08) × 10 ⁻³		885
$e^- \bar{\nu}_e \nu_\tau$	[g] (17.82 ± 0.04) %		888
$e^- \bar{\nu}_e \nu_\tau \gamma$	[e] (1.83 ± 0.05) %		888
$h^- \geq 0 K_L^0 \nu_\tau$	(12.03 ± 0.05) %		883
$h^- \nu_\tau$	(11.51 ± 0.05) %		883
$\pi^- \nu_\tau$	[g] (10.82 ± 0.05) %		883
$K^- \nu_\tau$	[g] (6.96 ± 0.10) × 10 ⁻³		820
$h^- \geq 1$ neutrals ν_τ	(37.01 ± 0.09) %		-
$h^- \geq 1 \pi^0 \nu_\tau$ (ex. K^0)	(36.51 ± 0.09) %		-
$h^- \pi^0 \nu_\tau$	(25.93 ± 0.09) %		878
$\pi^- \pi^0 \nu_\tau$	[g] (25.49 ± 0.09) %		878
$\pi^- \pi^0$ non-ρ(770) ν_τ	(3.0 ± 3.2) × 10 ⁻³		878
$K^- \pi^0 \nu_\tau$	[g] (4.33 ± 0.15) × 10 ⁻³		814
$h^- \geq 2 \pi^0 \nu_\tau$	(10.81 ± 0.09) %		-
$h^- 2 \pi^0 \nu_\tau$	(9.48 ± 0.10) %		862
$h^- 2 \pi^0 \nu_\tau$ (ex. K^0)	(9.32 ± 0.10) %		862
$\pi^- 2 \pi^0 \nu_\tau$ (ex. K^0)	[g] (9.26 ± 0.10) %		862
$\pi^- 2 \pi^0 \nu_\tau$ (ex. K^0)	< 9 × 10 ⁻³ CL=95%		862

$$\frac{\Gamma(\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau)}{\Gamma_{\text{tot}}} = \text{BF}(\tau^- \rightarrow e^- \nu \nu)$$

$$\Gamma_{\text{tot}} = \frac{1}{\tau} \Rightarrow \text{BF}(\tau^- \rightarrow e^- \nu \nu) = \tau_e \Gamma(\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau)$$

$$\text{BF}(\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau) = \frac{\#(\tau^- \rightarrow e^-)}{\# \text{total } \tau^- \text{ produced}}$$

$$\frac{\Gamma(\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu)}{\Gamma(\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau)} = \frac{\tau_\tau}{\tau_\mu} \frac{\text{BF}(\mu^- \rightarrow e^-)}{\text{BF}(\tau^- \rightarrow e^-)}$$